NORTHBRIDGE INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

B.TECH COMPUTER SCIENCE CURRICULUM 2026

Document code: NIT/CSE/CUR/2026
Version: Version 1.0
Effective from: 1 August 2026
Classification: Academic policy - approved for student reference

1. Programme Overview

1.1 The Bachelor of Technology in Computer Science and Engineering is a four-year undergraduate programme organised over eight semesters. The curriculum combines computing foundations, systems, software development, mathematical preparation, electives, internship and project work.

1.2 A student must earn at least 160 credits and satisfy all academic regulations to qualify for the degree. The normal duration is four years and the maximum permitted duration is seven years from initial registration.

2. Credit Structure

a. Computer science core courses and laboratories: 74 credits.

b. Mathematics, basic sciences, engineering and humanities: 34 credits.

c. Programme electives: 18 credits.

d. Open electives: 14 credits.

e. Internship and project work: 20 credits.

2.1 The semester plan carries 20 credits in each semester, for a programme total of 160 credits. Approved substitutions must preserve the applicable category and total credit requirement.

3. Semester I - 20 Credits

CS101 - Introduction to Programming (4 credits)

CS102 - Discrete Mathematics (3 credits)

MA101 - Engineering Mathematics I (4 credits)

PH101 - Engineering Physics (4 credits)

EE101 - Basic Electrical Engineering (3 credits)

HS101 - Communication Skills (2 credits)

4. Semester II - 20 Credits

CS202 - Object-Oriented Programming (4 credits). Prerequisite: CS101

CS203 - Computer Organization (4 credits)

MA102 - Engineering Mathematics II (4 credits)

EC101 - Digital Logic (3 credits)

ME101 - Engineering Design (3 credits)

HS102 - Professional Ethics (2 credits)

5. Semester III - 20 Credits

CS201 - Data Structures (4 credits). Prerequisite: CS101

CS204 - Database Foundations (4 credits)

MA201 - Linear Algebra and Probability (4 credits). Prerequisite: MA101

CS205 - Web Technologies (3 credits). Prerequisite: CS101

CS206 - Data Structures Laboratory (2 credits). Corequisite: CS201

OE201 - Open Elective I (3 credits)

6. Semester IV - 20 Credits

CS301 - Database Management Systems (4 credits). Prerequisites: CS201 and CS204

CS302 - Operating Systems (4 credits). Prerequisites: CS201 and CS203

CS303 - Computer Networks (4 credits). Prerequisite: CS203

CS304 - Design and Analysis of Algorithms (4 credits). Prerequisites: CS201 and MA102

CS307 - Systems Laboratory (2 credits). Corequisites: CS302 and CS303

HS201 - Economics for Engineers (2 credits)

7. Semester V - 20 Credits

CS305 - Software Engineering (4 credits). Prerequisite: CS202

CS401 - Artificial Intelligence (4 credits). Prerequisites: CS201 and MA201

CS402 - Information Security (4 credits). Prerequisites: CS302 and CS303

CS306 - Software Engineering Laboratory (2 credits). Corequisite: CS305

PE301 - Programme Elective I (3 credits)

OE301 - Open Elective II (3 credits)

8. Semester VI - 20 Credits

CS403 - Cloud Computing (4 credits). Prerequisites: CS302 and CS303

CS404 - Natural Language Processing (4 credits). Prerequisites: CS201 and MA201

CS405 - Machine Learning (4 credits). Prerequisites: CS301 and MA201

CS406 - Artificial Intelligence and Machine Learning Laboratory (2 credits). Corequisites: CS401 and CS405

PE302 - Programme Elective II (3 credits)

HS301 - Research Methods (3 credits)

9. Semester VII - 20 Credits

CS491 - Industry Internship (4 credits). Minimum approved duration: six weeks

CS492 - Project Phase I (6 credits)

PE401 - Programme Elective III (3 credits)

PE402 - Programme Elective IV (3 credits)

OE401 - Open Elective III (4 credits)

10. Semester VIII - 20 Credits

CS493 - Project Phase II (10 credits). Prerequisite: CS492

PE403 - Programme Elective V (3 credits)

PE404 - Programme Elective VI (3 credits)

OE402 - Open Elective IV (4 credits)

11. Prerequisite and Registration Rules

11.1 A prerequisite is satisfied only after the student earns a passing grade of P or higher in the stated course. A corequisite must be completed earlier or registered in the same semester.

11.2 Students should follow the semester sequence shown above. Registration outside the sequence requires academic advising, prerequisite compliance and an available timetable.

11.3 The normal semester load is 18 to 24 credits. A load up to 27 credits requires approval. A student with a CGPA below 5.00 is ordinarily limited to 20 credits.

12. Electives

12.1 Programme electives provide advanced study within computing. The department announces the available list before registration. An elective is offered only when staffing, facilities and minimum enrolment requirements are met.

12.2 Indicative programme electives include Data Mining, Distributed Systems, Computer Vision, Advanced Databases, Mobile Computing, DevOps Engineering, Cryptography and Human-Computer Interaction.

12.3 Open electives may be selected from approved offerings outside the home programme. A course already counted toward another curriculum category cannot be counted again as an open elective.

13. Internship and Project Work

13.1 CS491 requires an approved internship of at least six weeks. Students must obtain departmental approval before joining the host organisation and must submit a report and supervisor assessment after completion.

13.2 Project Phase I covers problem definition, literature review, requirements, method selection and an initial implementation. Project Phase II continues the approved work and concludes with a dissertation, demonstration and oral examination.

13.3 Registration for final-year project work normally requires at least 108 earned credits and no more than four active backlogs. The department may require completion of specific supporting courses for a proposed topic.

14. Progression and Degree Requirements

14.1 Progression to the third year normally requires at least 70 earned credits. Progression to the fourth year normally requires at least 108 earned credits and no more than four active backlogs.

14.2 For graduation, a student must earn 160 credits, complete all compulsory components, achieve a CGPA of at least 5.00 and clear all outstanding academic or disciplinary requirements.

14.3 Failure in a course does not remove its prerequisite effect. The student must pass the course before registering for a dependent course unless an explicit curriculum transition rule states otherwise.

15. Curriculum Transition and Interpretation

15.1 This curriculum applies to students admitted from the 2026 cohort. Students from an earlier cohort continue under their approved curriculum unless formally transferred to this version.

15.2 When a course is discontinued, the department may approve an equivalent replacement and publish the mapping. Credits are counted once, using the approved category of the replacement course.

15.3 The Board of Studies may revise elective lists, course content and prerequisite mappings without changing the degree requirements stated here. Any revision takes effect only after an official notice is issued.